

Ameet Rao's Calculus III Notes

Quadric Surfaces: Ellipsoids, Hyperboloids, Cones, and Paraboloids

Quadric Surfaces

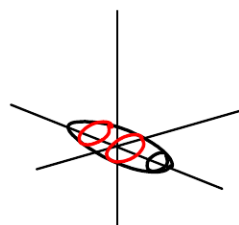
3D Conic Sections

Quadric surfaces are the 3D versions of conic sections. Identify one by its equation, then confirm with its traces (slices by the coordinate planes).

Ellipsoid

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

If $a=b=c=1 \rightarrow$ sphere



Trace

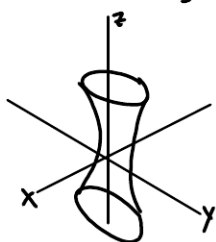
xy
 xz
 yz

} ellipse

Ellipsoid: all squared terms positive, equal to 1. If $a = b = c$ it is a sphere. Every trace is an ellipse.

Hyperboloid of one sheet

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$$



Trace

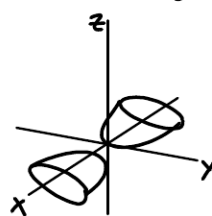
xy ellipse
 xz
 yz Hyperbola

Neg term determines direction

Hyperboloid of one sheet: two positive terms, one negative. The negative term names the axis it wraps around. Slices across that axis are ellipses; slices along it are hyperbolas.

Hyperboloid of two sheets

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$$



Trace

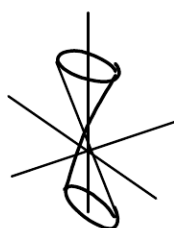
xy Hyperbola
 xz Hyperbola
 yz ellipse

pos term determines direction

Hyperboloid of two sheets: one positive term, two negative. The positive term names the axis along which the two pieces open. Slices along that axis are hyperbolas.

Elliptic Cone

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0$$



Neg term determine direction

Trace

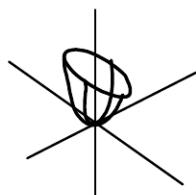
xy ellipse
 xz
 yz Hyperbola

Elliptic cone: same form as the one-sheet hyperboloid but set equal to 0. The negative term names the axis. Horizontal slices are ellipses; vertical slices are hyperbolas.

Elliptic paraboloid

$$z = \frac{x^2}{a^2} + \frac{y^2}{b^2}$$

Non squared term determine direction



Trace

xy ellipse
 xz parabola
 yz parabola

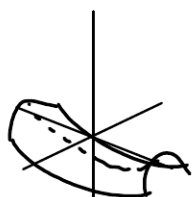
Elliptic paraboloid: one variable is not squared, and it names the axis the bowl opens along. Slices across the axis are ellipses; slices along it are parabolas.

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Hyperbolic Paraboloid (pringle)

$$z = \frac{y^2}{b^2} - \frac{x^2}{a^2}$$



Trace
xy Hyperbola
xz parabola
yz parabola

Hyperbolic paraboloid (the "pringle" saddle): one squared term is positive, one negative, and the third variable is not squared. Traces are hyperbolas and parabolas.

① $4x^2 - 3y^2 + 12z^2 + 12 = 0$

② $x - y^2 - 4z^2 = 0$

③ $x^2 + 2y^2 + z^2 - 4x + 4y - 2z + 3 = 0$

Classify by rearranging. ① Divide by -12 : one positive term, so a two-sheet hyperboloid. ② $x = y^2 + 4z^2$: an elliptic paraboloid. ③ Complete the square: $(x-2)^2 + 2(y+1)^2 + (z-1)^2 = 4$, an ellipsoid.